



A common passive probe (Credit: Rhode and Schwarz)

tached in order to monitor a signal. Only when the probe provides the oscilloscope with the right inputs can it provide accurate measurements. Make sure to choose an oscilloscope that offers all the special probes you need for your application.

Newer parameters

Oscilloscopes have evolved over the years, and it is hard to believe that this technology is over 100 years old! Therefore, it is important to consider some more parameters, especially if you are upgrading your device from a very old version.

Quality of the display. The display of the oscilloscope shows the amplitude (usually voltage) of a signal on the Y-axis and time along the X-axis. Choose an oscilloscope with a range of trace intensity gradations so you may detect small signal irregularities and features in the waveform. The brightness (intensity) of a waveform on a display indicates how frequently a signal appears at that specific location on the display. This means that intensity is an important parameter you should consider. Also, it is simpler to display numerous measurements and locate problematic transient signals on large, high-resolution screens.

Application support. Application software for optical and electrical design troubleshooting and standard compliance is available on advanced scopes. Additionally, oscilloscopes need to include sup-

port for debugging embedded systems that use CAN/LIN, PC, SPI, FlexRay, MOST, and other mixed, analogue and digital, parallel and serial technologies. You should be able to easily view signals in the frequency domain as well, especially if you are working

on RF circuits. So, make sure the oscilloscope provides you with the necessary application support.

Interfaces and Connectivity. It is always wise to choose an oscilloscope that can be quickly connected to other devices for advanced features and has less complicated interfacing documentation. Furthermore, you should consider an oscilloscope that enables you to connect to a Windows or Linux system for file sharing. Also check if your selected oscilloscope has all the standard interfaces, such as GPIB, USB, RS-232, Ethernet, and LXI.

Powerful analysis. Tools such as zoom, panning, pause, and play are very crucial as they help you focus on what's important. Moreover, some oscilloscopes have the 'mark' feature, which allows you to mark events of importance that you may like to review later. Some models include advanced search capabilities that let you define search criteria to instantly identify every occurrence

The Rule of 5

Most vendors recommend the '5 Times Rule' while selecting an oscilloscope:

- The sampling rate of the oscilloscope should be at least five times the highest frequency.
- The rise time of the oscilloscope should be five times faster than the fastest signal.
- The oscilloscope's bandwidth should be five times that of the measurement's fastest signal.

in an acquisition that does not meet some specified criterion, such as setup and hold time.

Remote usage. Remote access to an oscilloscope's data has become essential as many companies operate in geographically separated locations. Controlling a scope physically from a computer is one approach. Here, a virtual scope front panel provided by a GUI loaded on the PC is used to control the scope, which is connected to the PC through a network. Another approach is saving the waveform data in a file format that may be downloaded from the oscilloscope to a PC where specialised software shows the captured waveform. For example, Tektronix offers the TekScope PC Software for this purpose. Do check if your preferred vendor has such options.

General parameters

Whether you are buying a new phone, car, or any electronic device, there are certain things that are always there on your mind. Is the company reliable? Is the device user-friendly? The same thing is true when it comes to buying oscilloscopes. Here are some general parameters that you should look out for.

Guaranteed specs and good documentation. When selecting an oscilloscope, you should make sure its datasheet lists all the parameters you need to measure as 'guaranteed specifications.' The 'typical' parameters are merely oscilloscope performance indicators and cannot be used to obtain accurate readings that meet established quality standards.

An oscilloscope's technical parameters can easily cover many pages. Make sure that your selected oscilloscope's datasheet has all the important specifications listed without any ambiguity. There are typically a lot of footnotes on the datasheet that describe how a standard was created and/or measured in